EFY TESTED S.C. Dwivedi

Hot Water Utilisation Enhancer For Solar Water Heater

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The circuit presented here is a simple device to enhance the utilisation of hot water from a solar water heater system. Normally, during day time, you get hot water continuously from the solar water heater due to the presence of sunlight. But in absence of sunlight, freshwater keeps getting added to the tank and cools the water, which is not desirable, especially in winter. This is not a good way of utilising hot water at home.

By controlling cold water at the tank inlet in the absence of sunlight, we can utilise the available hot water in a better way. This circuit is specially designed to fulfil this requirement and can be installed in the existing solar water heater by making some minor changes.

The typical block diagram of the solar water heater system is shown in Fig. 1. Hot water gets circulated through a solar collector, which collects heat by absorbing sunlight. This water is then stored in a storage tank, or piped into an additional backup water heater tank. We are going to add a solenoid valve at cold water inlet in such a solar water heater system.

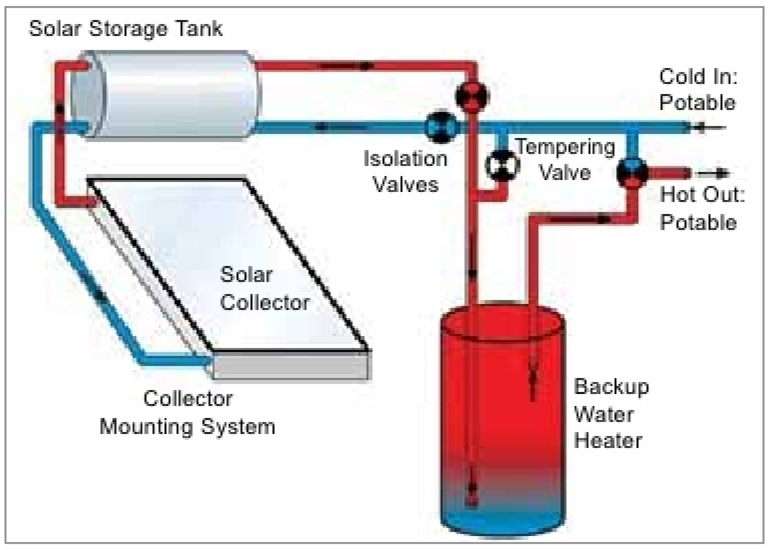

Fig. 1: Block diagram of solar water heater system (Credit: cleangreenenergyzone.com)

Circuit and working

The circuit diagram of the hot water utilisation enhancer is shown in Fig. 2. It consists of Arduino Uno (Board1), LDR1 (light dependent resistor), 12V single changeover (1CO) relay, 12V solenoid valve, and a few other components. The circuit diagram of the power supply is shown in Fig. 3.

Arduino Uno. Arduino Uno board is connected to LDR1, indicator LED1, and relay RL1. The light intensity read by LDR1 is recorded and processed by the Arduino Uno board, which further commands the relay according to the program's condition.

Power supply. The power supply is used to power a 12V solenoid valve and relay (RL1) for switching. 5V DC is also derived from this power supply. It consists of a 230V primary to 15V AC, 1A secondary transformer (X1). The output of this step-down transformer is given to full-bridge rectifier BR1 and two capacitors of value 1000µF, 35V (C1), and 1µF, 25V (C2). 12V regulator 7812 (IC1) and 5V regulator 7805 (IC2) are used for providing 12V and 5V regulated power supplies, respectively. Diodes D2 and D3 are provided to prevent reverse polarity during connections. LED2 is a 12V DC indicator, and LED3 is a 5V DC indicator.

Light-dependent resistor. It is connected to analogue pin A0 of the Arduino board. LDR1 is used as a

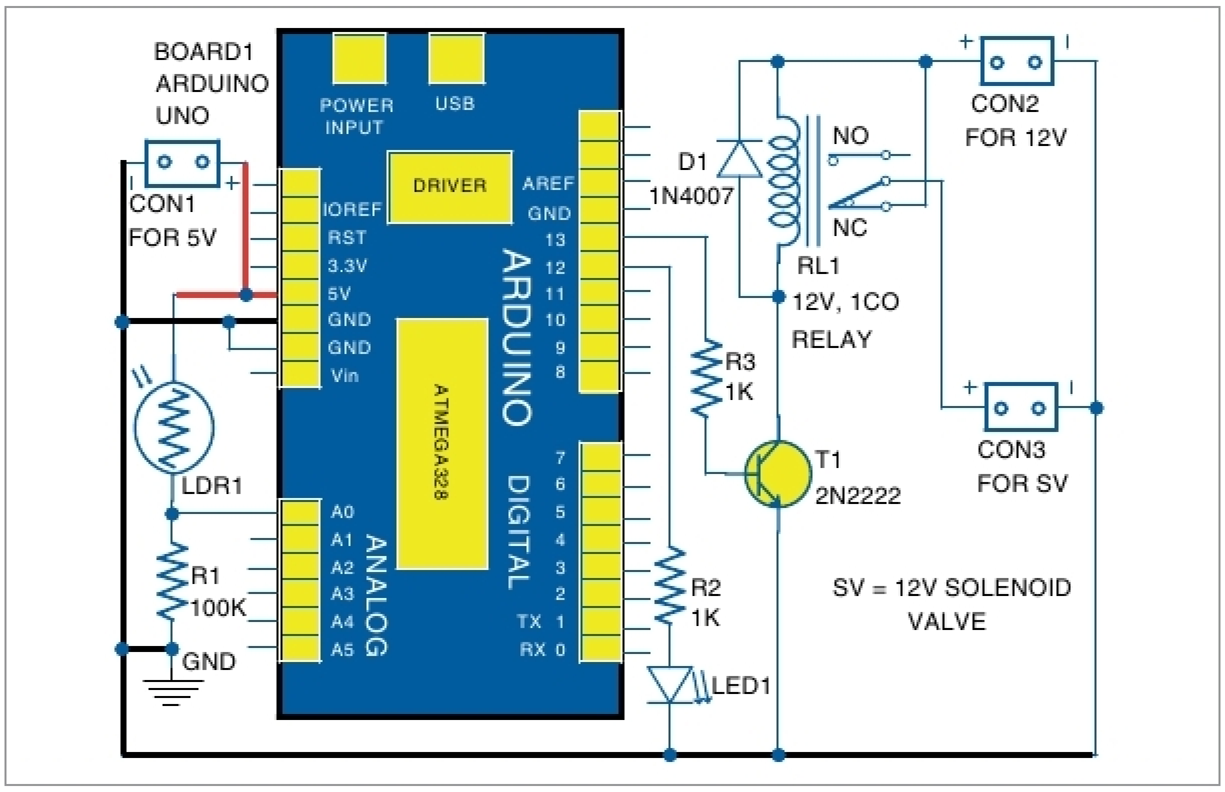

Fig. 2: Circuit diagram of hot water utilisation enhancer for solar water

PARTS LIST

Semiconductors:	
Board1	- Arduino Uno
IC1	- 7812, 12V voltage regulator
IC2	- 7805, 5V voltage regulator
D1-D3	- 1N4007 rectifier diode
LED1-LED3	- 5mm LED
T1	- 2N2222 npn transistor
BR1	- 1A bridge rectifier
Resistors (all 1/4-watt, ±5% carbon):	
R1	- 100-kilo-ohm
R2, R3	- 1-kilo-ohm
R4	- 2.2-kilo-ohm
R5	- 220-ohm
Capacitors:	
C1	- 1000µF, 35V electrolytic
C2	- 1µF, 25V electrolytic
C3	- 0.22µF ceramic disk
C4	- 0.1µF ceramic disk
Miscellaneous:	
CON1-CON5	- 2-pin connector
X1	- 230V AC primary to 15V, 1A secondary transformer
RL1	- 12V, single changeover relay
SV	- 2V solenoid valve normally closed (NC) type